Binary -- Guided Notes ANSWER KEY

Unit tp-1.2 | CompTIA Network+ N10-009 Obj. FC0-U71 1.2 | N10-008 FC0-U71 1.2

For instructor use / distribute after students complete the notes.

Question 1

Binary is a base-_______ number system that uses only the digits _______ and _______. Each position in a binary number represents a power of two. The rightmost position represents 2^0 = _______, and the position immediately to its left represents 2^1 = _______.

Answer: Base-2; 0 and 1; 1; 2.

Question 2

Fill in the 8 bit-position values from left (most significant) to right (least significant):
___ | ___ | ___ | ___ | ___ | ___ | ___ | ___

Answer: 128 | 64 | 32 | 16 | 8 | 4 | 2 | 1.

Question 3

Convert the binary number 10110010 to decimal. Show your work by listing which bit positions are set to 1 and adding their values.

Answer: Positions set to 1: 128, 32, 16, 2. Sum: 128 + 32 + 16 + 2 = 178.

Question 4

Convert the decimal number 205 to binary using the subtraction method. Start at 128 and work right. Show each step.

Answer: 128 fits (205 - 128 = 77). 64 fits (77 - 64 = 13). 32 does not fit. 16 does not fit. 8 fits (13 - 8 = 5). 4 fits (5 - 4 = 1). 2 does not fit. 1 fits. Result: 11001101.

Question 5

One byte equals _______ bits. One nibble equals _______ bits. A byte can represent values from 0 to a maximum of _______. The maximum value occurs when all 8 bits are set to _______.

Answer: 8 bits; 4 bits; 255; 1 (all ones: 11111111).

Question 6

The MSB stands for _______ and is the _______ (leftmost/rightmost) bit in a binary number. It has the _______ value. The LSB stands for _______ and is the _______ (leftmost/rightmost) bit, with value _______.

Answer: Most Significant Bit; leftmost; highest (128 in a byte). Least Significant Bit; rightmost; 1.

Question 7

Computers use binary because transistors -- the basic building blocks of processors -- have exactly _______ stable states: fully _______ (conducting, = 1) and fully _______ (not conducting, = 0). This makes binary arithmetic reliable and easy to implement in _______.

Answer: Two; on; off; hardware (silicon).

Question 8 -- Real World Application

REAL WORLD: A student claims that the binary number 11111111 is the same as the decimal number 255. Another student says it must be more because there are eight 1s. Explain who is correct and why, showing the math using bit position values.

Answer: The first student is correct. To convert 11111111 to decimal, add all bit position values where a 1 appears: 128 + 64 + 32 + 16 + 8 + 4 + 2 + 1 = 255. The second student is thinking of the digits as independent numbers rather than positional values. In binary, each digit represents a power of two based on its position -- not its face value. Eight 1-bits does not mean 8; it means the sum of all 8 powers of two from 2^0 through 2^7, which equals 255.